

Lecture 05

Principal Component Analysis

Principal Components Analysis (PCA)

- PCA is a useful statistical technique for finding patterns in data of high dimension.
- It is used to express the data in such a way as to highlight their similarities and differences.

Background Statistics

1 → Mean

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$$

1 → Standard Deviation

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{(n-1)}}$$

1 → Variance

$$s^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{(n-1)}$$

Covariance

→ Standard deviation and variance can only operate on one dimensional data.

→ Covariance is a measure of how much two dimensions vary from the mean with respect to each other.

$$\text{var}(X) = \frac{\sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})}{(n - 1)}$$

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{(n - 1)}$$

Covariance: An Example

	<i>Hours(H)</i>	<i>Mark(M)</i>
Data	9	39
	15	56
	25	93
	14	61
	10	50
	18	75
	0	32
	16	85
	5	42
	19	70
	16	66
	20	80
Totals	167	749
Averages	13.92	62.42

<i>H</i>	<i>M</i>	$(H_i - \bar{H})$	$(M_i - \bar{M})$	$(H_i - \bar{H})(M_i - \bar{M})$
9	39	-4.92	-23.42	115.23
15	56	1.08	-6.42	-6.93
25	93	11.08	30.58	338.83
14	61	0.08	-1.42	-0.11
10	50	-3.92	-12.42	48.69
18	75	4.08	12.58	51.33
0	32	-13.92	-30.42	423.45
16	85	2.08	22.58	46.97
5	42	-8.92	-20.42	182.15
19	70	5.08	7.58	38.51
16	66	2.08	3.58	7.45
20	80	6.08	17.58	106.89
Total				1149.89
Average				104.54

→ **Positive value** indicates both dimensions increase together.

→ **Negative value** indicates as one increases, the other decreases.

→ **Zero** indicates dimensions are independent.

Covariance matrix

$$C = \begin{pmatrix} cov(x, x) & cov(x, y) & cov(x, z) \\ cov(y, x) & cov(y, y) & cov(y, z) \\ cov(z, x) & cov(z, y) & cov(z, z) \end{pmatrix}$$

Eigenvector and Eigenvalue

→ An eigenvector or characteristic vector of a square matrix is a vector that does not change its direction under the associated linear transformation.

→ If $\mathbf{v}$ is a vector that is not zero, then it is an eigenvector of a square matrix A if $A\mathbf{v}$ is a scalar multiple of $\mathbf{v}$. This condition could be written as the equation:

$$A\mathbf{v} = \lambda\mathbf{v} \text{ where } \lambda \text{ is the eigenvalue}$$

An Example

→ Find the eigenvector and eigenvalue of the following matrix.

$$A = \begin{pmatrix} 7 & 3 \\ 3 & -1 \end{pmatrix}$$

Eigenvector and Eigenvalue

Def'n Let A be an $n \times n$ matrix. A scalar λ is called an eigenvalue of A if there is a nonzero vector $\bar{x}$ such that $A\bar{x} = \lambda\bar{x}$. Such a vector $\bar{x}$ is called an eigenvector of A corresponding to λ .

Note: if λ is an eigenvalue of A , and $\bar{x}$ is an eigenvector belonging to λ , any nonzero multiple of $\bar{x}$ will be an eigenvector.

Assuming: $A\bar{x} = \lambda\bar{x}$

$$\underline{A(\alpha\bar{x})} = \alpha \underline{A\bar{x}} = \alpha \lambda\bar{x} = \underline{\lambda(\alpha\bar{x})}$$

Finding eigenvector and eigenvalue

To solve for the eigenvalues, λ_i , and the corresponding eigenvectors, $\bar{x}_i$, of an $n \times n$ matrix A , do the following:

- ① Multiply an $n \times n$ identity matrix by the scalar λ
- ② Subtract the identity matrix multiple from the matrix A
- ③ Find the determinant of the matrix and the difference
- ④ Solve for the values of λ that satisfy the equation $\det(A - \lambda I) = 0$
- ⑤ Solve for the corresponding vector to each λ

Finding eigenvector and eigenvalue

$$\textcircled{1} \quad \lambda I = \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$\begin{aligned} \textcircled{2} \quad A - \lambda I &= \begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \\ &= \begin{bmatrix} 7-\lambda & 3 \\ 3 & -1-\lambda \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad \det \begin{bmatrix} 7-\lambda & 3 \\ 3 & -1-\lambda \end{bmatrix} \\ &= (7-\lambda)(-1-\lambda) - (3)(3) \\ &= -7 - 7\lambda + \lambda + \lambda^2 - 9 \\ &= \lambda^2 - 6\lambda - 16 \end{aligned}$$

Finding eigenvector and eigenvalue

$$\lambda^2 - 6\lambda - 16 = 0$$

$$(\lambda - 8)(\lambda + 2) = 0$$

$$\lambda = 8$$

$$\lambda = -2$$

$$\textcircled{5} \begin{bmatrix} 7-\lambda & 3 \\ 3 & -1-\lambda \end{bmatrix}$$

$$\lambda = 8:$$

$$\begin{bmatrix} 7-8 & 3 \\ 3 & -1-8 \end{bmatrix} = \begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix}$$

$\underbrace{\hspace{10em}}_B$

Finding eigenvector and eigenvalue

Solve: $B\bar{x} = \bar{0}$

$$\begin{bmatrix} -1 & 3 \\ 3 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
$$\left[\begin{array}{cc|c} -1 & 3 & 0 \\ 3 & -9 & 0 \end{array} \right] \xrightarrow{3R_1 + R_2 \rightarrow R_2} \left[\begin{array}{cc|c} -1 & 3 & 0 \\ 0 & 0 & 0 \end{array} \right]$$

$$\begin{aligned} -x_1 + 3x_2 &= 0 \\ 3x_2 &= x_1 \\ \text{Let } \boxed{x_2 = 1} \\ 3(1) &= \boxed{x_1 = 3} \end{aligned}$$

Eigen Vector: $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$

Finding eigenvector and eigenvalue

check: $\begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 8 \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} 24 \\ 8 \end{bmatrix} = \begin{bmatrix} 24 \\ 8 \end{bmatrix}$$

$$\lambda = -2$$
$$\begin{bmatrix} 7 - (-2) & 3 \\ 3 & -1 - (-2) \end{bmatrix} = \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}$$
$$\left[\begin{array}{cc|c} 9 & 3 & 0 \\ 3 & 1 & 0 \end{array} \right] \rightarrow \begin{bmatrix} x_1 & x_2 \\ 0 & 0 & 0 \\ 3 & 1 & 0 \end{bmatrix}$$

$-3R_2 + R_1 \rightarrow R_1$

Finding eigenvector and eigenvalue

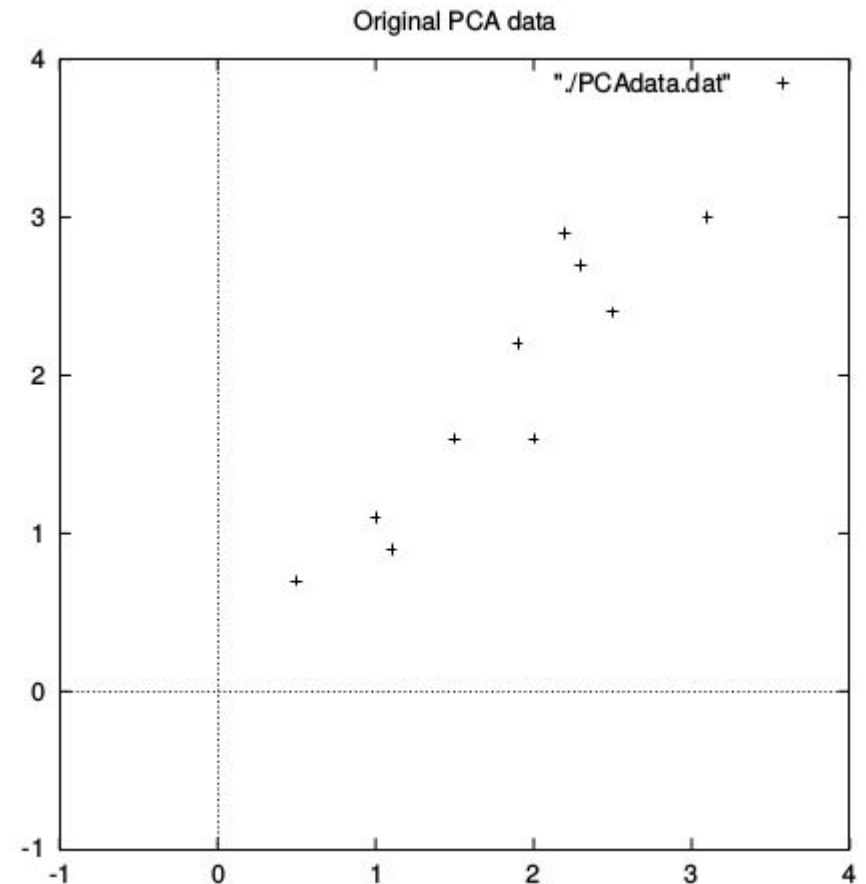
$$\begin{aligned}3x_1 + x_2 &= 0 \\x_2 &= -3x_1 \\ \text{let } x_1 &= 1 \\x_2 &= -3 \\ \text{Eigenvector: } &\begin{bmatrix} 1 \\ -3 \end{bmatrix} \\ \text{eigenvalue: } &-2\end{aligned}$$

Steps of PCA

- Step 1: Get some data
- Step 2: Subtract the mean
- Step 3: Calculate the covariance matrix
- Step 4: Calculate eigenvectors and eigenvalues of covariance matrix
- Step 5: Choosing components and forming a feature vector

Dataset and subtracting the mean

Data =	x	y	DataAdjust =	x	y
	2.5	2.4		.69	.49
	0.5	0.7		-1.31	-1.21
	2.2	2.9		.39	.99
	1.9	2.2		.09	.29
	3.1	3.0		1.29	1.09
	2.3	2.7		.49	.79
	2	1.6		.19	-.31
	1	1.1		-.81	-.81
	1.5	1.6		-.31	-.31
	1.1	0.9		-.71	-1.01



Calculating the Covariance matrix

$$cov = \begin{pmatrix} .616555556 & .615444444 \\ .615444444 & .716555556 \end{pmatrix}$$

$$cov(X, Y) = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{(n - 1)}$$

Eigenvalues and Eigenvectors

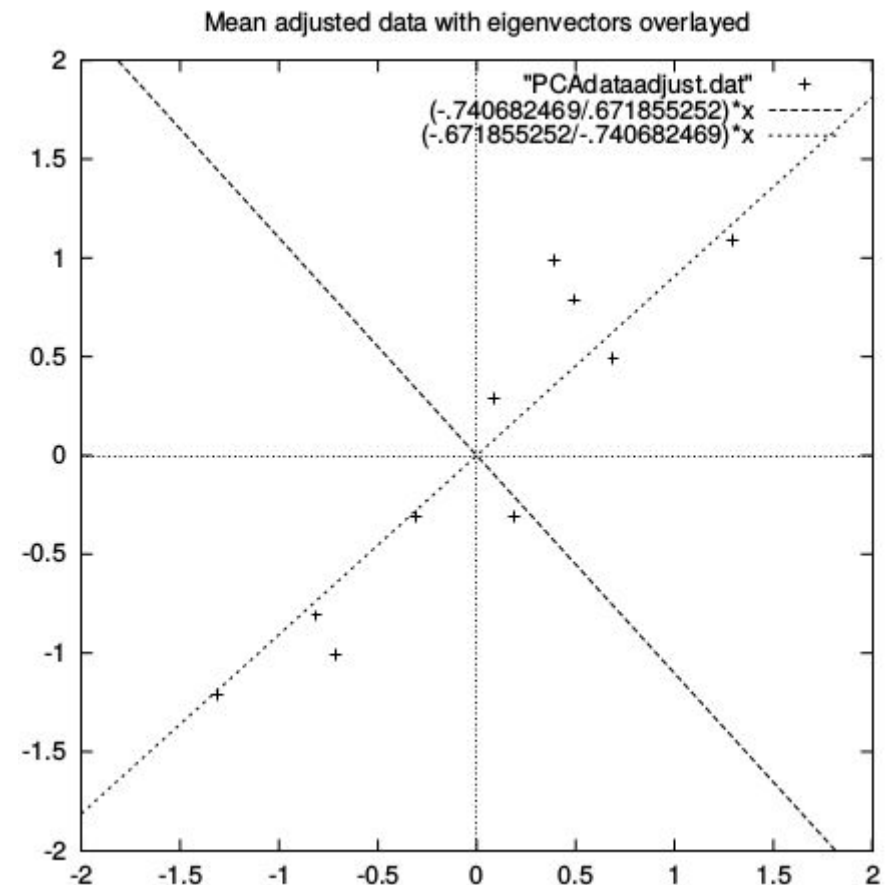
$$\text{eigenvalues} = \begin{pmatrix} .0490833989 \\ 1.28402771 \end{pmatrix}$$

$$\text{eigenvectors} = \begin{pmatrix} -.735178656 & -.677873399 \\ .677873399 & -.735178656 \end{pmatrix}$$

Eigenvalues and Eigenvectors

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Forming a feature vector

- The eigenvector with the highest eigenvalue is the principle component of the data set.
- Sort the eigenvalues in descending order.
- From n eigenvalues, take top p eigenvalues.
- Dimensions will be changed from n to p .

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$$\text{FeatureVector} = (eig_1 \ eig_2 \ eig_3 \ \dots \ eig_n)$$

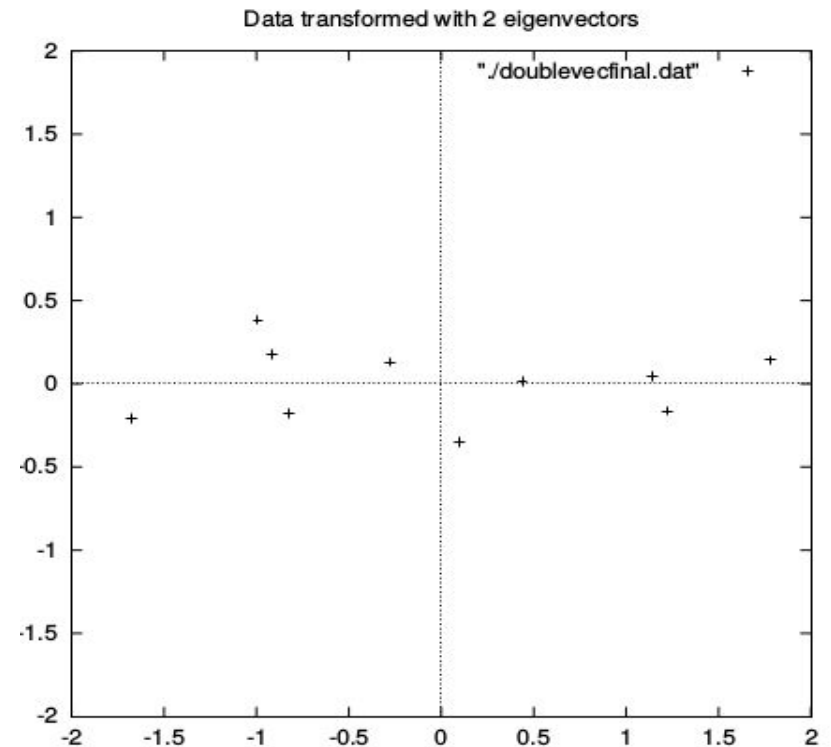
Deriving new dataset

→ Take the transpose of the vector and multiply it on the left of the original data set, transposed.

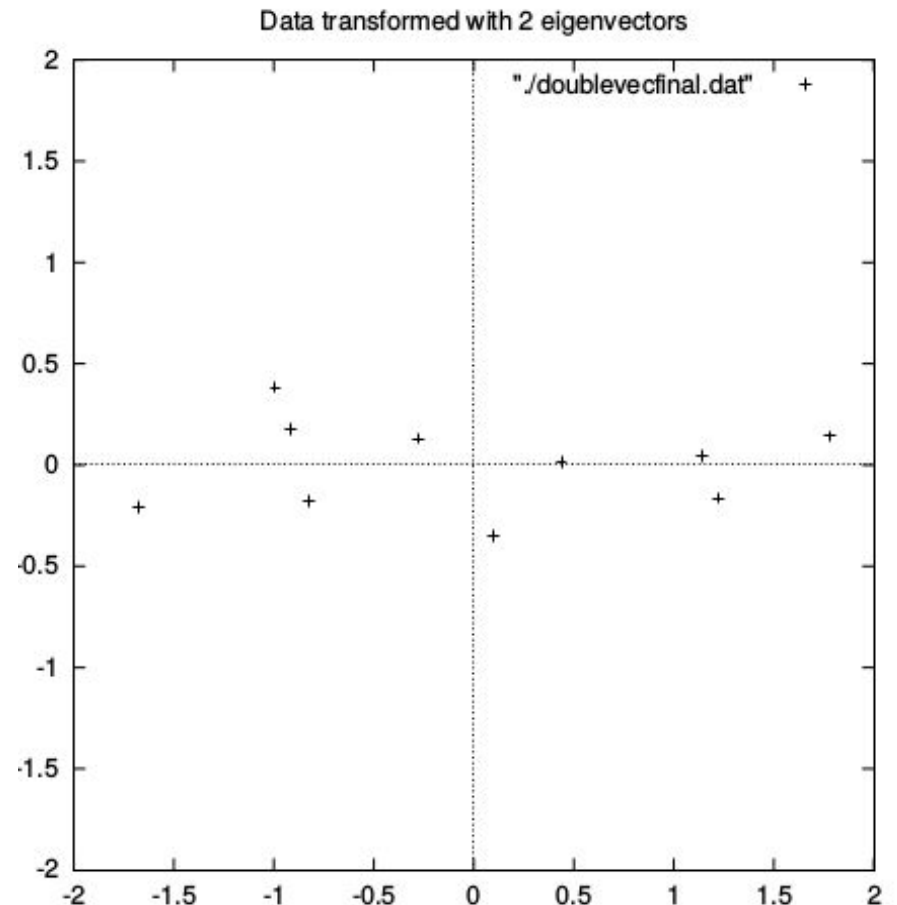
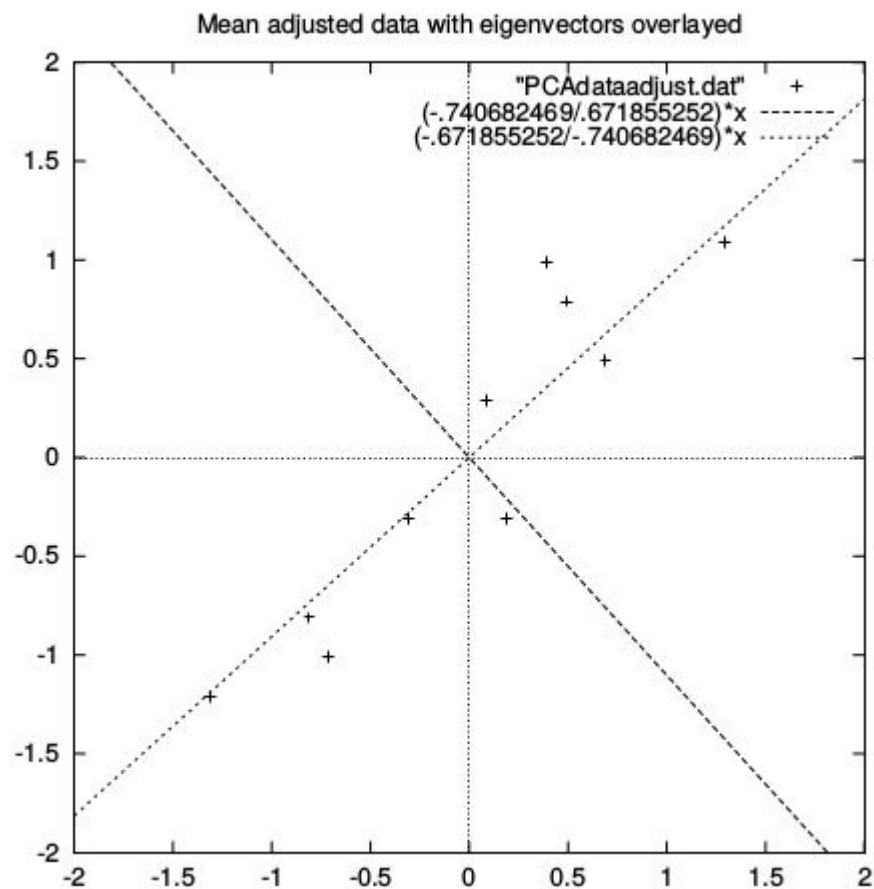
$$\text{FinalData} = \text{RowFeatureVector} \times \text{RowDataAdjust},$$

Transformed Data=

x	y
-.827970186	-.175115307
1.77758033	.142857227
-.992197494	.384374989
-.274210416	.130417207
-1.67580142	-.209498461
-.912949103	.175282444
.0991094375	-.349824698
1.14457216	.0464172582
.438046137	.0177646297
1.22382056	-.162675287



Old and new dataset



Getting Original Data back

→ Getting back exact original data is only possible if we take all eigenvectors during transformation. Otherwise, we lose some information.

$$FinalData = RowFeatureVector \times RowDataAdjust,$$

$$RowDataAdjust = RowFeatureVector^T \times FinalData$$

$$RowOriginalData = (RowFeatureVector^T \times FinalData) + OriginalMean$$